Beyond Forest Loss: Identifying Deforestation Archetypes

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https://github.com/ssrakash5/deforestation-archetypes — Dataset: Google Drive

Abstract

Global forest monitoring systems reliably detect where and when deforestation occurs, yet their binary outputs—loss versus no-loss—cannot distinguish road construction from agricultural clearing or selective logging. We present an unsupervised pipeline that discovers repeatable spatial *archetypes* directly from event morphology, requiring no ground-truth driver labels. Applied to the SpaceNet-7 dataset (1,381 change events, 14 morphological descriptors), UMAP + HDBSCAN recovers **14 stable archetypes** (noise rate 5.6 %). Cluster stability across five randomised seeds yields a **mean ARI of 0.658** (range 0.56–0.78). A k-Nearest Neighbour classifier achieves **98.24 % five-fold cross-validation accuracy** on held-out events. A logistic proxy risk model combining area, irregularity, and perimeter reaches **ROC-AUC = 0.994**, with certain archetypes exhibiting near-100 % high-risk fractions. The pipeline generalises to a 15,000-event Hansen Global Forest Change sample without modification, validating transferability at geographic scale.

Keywords: deforestation, unsupervised clustering, UMAP, HDBSCAN, morphological archetypes, remote sensing, forest monitoring.

I. INTRODUCTION

Tropical and subtropical forests continue to disappear at alarming rates, driving biodiversity loss, carbon release, and disrupted hydrological cycles. Systems such as the Hansen Global Forest Change product and Global Forest Watch (GFW) now provide annual 30-metre resolution loss maps at planetary scale—a remarkable monitoring achievement. Yet these outputs are fundamentally binary: a pixel is either forested or cleared. The underlying *process*—whether a road is being cut, a field is being prepared, or selective loggers are removing individual trees—remains invisible.

This distinction matters critically for conservation response. Road incursions require different enforcement strategies than agricultural conversion; fragmented selective logging has different long-term recovery trajectories than bulk clearing. If recurring spatial signatures of these processes could be identified automatically, monitoring systems could move from answering *where and when* forest loss occurred to answering *what kind and why*.

This is not only a scientific question but also an operational one. In many monitoring pipelines, analysts are overwhelmed by the number of alerts and must decide quickly which events deserve deeper inspection. A compact, contiguous clearing in an already converted landscape often calls for a different response than a newly emerging linear corridor penetrating previously intact forest. A useful intermediate representation should therefore be more expressive than a binary mask, but still simple enough to compute robustly at scale.

We propose framing deforestation characterisation as an *unsupervised archetype discovery* problem. Rather than imposing expert-defined categories, we let the geometric structure of binary change masks speak for itself. The central hypothesis is that different deforestation drivers leave distinct morphological fingerprints—elongated corridors for roads, compact polygons for agricultural conversion, fragmented mosaics for selective logging—and that these fingerprints are discoverable without any label supervision.

Our approach is intentionally conservative in its assumptions. We do not require hand-labeled driver annotations, and we do not assume that one globally correct taxonomy of deforestation types already exists. Instead, we ask a simpler and more robust question: do repeatable families of event geometry emerge from the data itself, and if they do, are they stable and useful enough to support prioritization and downstream classification?

The contributions of this work are:

1. A reproducible morphology feature extraction pipeline for binary change-mask events.
2. UMAP + HDBSCAN clustering recovering 14 stable deforestation archetypes.
3. Cluster stability analysis across randomised seeds (mean ARI 0.658, range 0.56–0.78).
4. A k-NN classifier enabling rapid archetype assignment (98.24 % CV accuracy).
5. A proxy risk model (ROC-AUC = 0.994) translating morphology into actionable risk scores.
6. Generalisation results on 15,000 Hansen events demonstrating geographic transferability.

II. RELATED WORK

Deforestation driver classification has been approached via expert digitisation [1], object-based image analysis [2], and supervised deep learning on high-resolution imagery [3]. These approaches require labelled training data that is expensive to collect at scale. Unsupervised clustering of land-cover change is less explored: Bullock et al. [4] cluster spectral trajectories rather than spatial geometry, while Qin et al. [5] apply k-means to patch statistics without density-based noise handling. HDBSCAN [6] and UMAP [7] have been applied together in other remote-sensing domains but not previously to deforestation event morphology. We fill this gap.

More broadly, this work sits at the intersection of two trends in remote sensing: first, a move from pixel-wise detection toward object- or event-level reasoning, and second, a renewed interest in interpretable unsupervised structure discovery. For forest monitoring, that combination is especially appealing. Event shapes can be summarized with a small number of descriptors that are easy to inspect, yet those descriptors often capture information that is directly meaningful to practitioners, such as elongation, fragmentation, and exposed edge.

III. DATA

A. Primary Dataset — SpaceNet-7

The SpaceNet-7 Multi-Temporal Urban Development dataset provides Planet Basemap imagery at 4 m/pixel across 101 scenes globally, with monthly acquisitions spanning 2017–2020. Binary change maps are derived from consecutive monthly acquisitions; connected-component extraction yields **1,381 individual event patches** across train and test splits. Each event is described by 542 features: 14 morphological shape descriptors and a 512-dimensional DINO ViT image embedding [10].

Although SpaceNet-7 was originally designed for change detection in the built environment, it offers a valuable testbed for the present study because it contains a large number of localized event masks with varied geometry. This lets us stress-test whether simple shape information is sufficient to reveal repeatable event families before scaling to a much noisier global forest-loss dataset.

B. Generalisation Dataset — Hansen GFC

The Hansen Global Forest Change v1.10 product [9] provides annual Landsat-derived 30-metre forest-loss maps spanning 2000–2022. A stratified geographic sample of **15,000 loss events** is extracted to test pipeline generalisation. Only the 14 morphological features are available for this dataset (no image embeddings), providing a strict transfer benchmark.

This second dataset serves an important role in the paper. It is larger, more heterogeneous, and more representative of operational forest monitoring than the SpaceNet benchmark. If the same pipeline continues to reveal structure under these conditions, then the archetypes are more likely to reflect genuine event morphology rather than benchmark-specific artifacts.

C. Feature Engineering

All features are computed deterministically from the binary event mask. Table 1 lists the 14 morphological descriptors.

These descriptors were chosen because they capture complementary aspects of shape. Area and perimeter measure event scale; compactness and solidity measure how filled-in or irregular the mask is; eccentricity and aspect ratio distinguish corridor-like events from block-like ones; and connected components or edge density help capture fragmentation. Together, they form a lightweight but interpretable representation that is well suited to unsupervised discovery.

An additional advantage of this representation is that it is easy to audit. If an archetype appears unusual, one can inspect the contributing descriptors and understand why it formed. This is a useful property in environmental monitoring, where users often need to justify why a given event was prioritized. In that sense, the feature space is not only compact but also communicable to non-ML stakeholders such as conservation practitioners or policy analysts.

Table 1: *Morphological features extracted per deforestation event.*

Feature	Description
`area_px`	Total mask area (pixels)
`perimeter_px`	Perimeter length (pixels)
`compactness`	$4\pi \cdot \text{area}/\text{perimeter}^2$
`eccentricity`	Focal distance / semi-major axis
`solidity`	area / convex hull area
`extent`	area / bounding-box area
`aspect_ratio`	Bounding-box width / height
`n_components`	No. of connected components
`edge_density`	Edge-pixel fraction within mask
`mask_frac`	Fraction of crop covered by mask
`crop_h/crop_w`	Event crop dimensions

IV. METHODS

Figure 1 summarises the full workflow. The key design choice is to keep archetype discovery unsupervised while still making the resulting representation useful for downstream monitoring and prioritisation.

A. Preprocessing

All 14 morphological features are standardised to zero mean and unit variance using `StandardScaler` [11]. Infinite values and NaN entries—arising from degenerate single-pixel events—are filled with zero prior to scaling.

This preprocessing step is important because the raw descriptors operate on very different numeric scales. Without standardisation, large-magnitude quantities such as area or perimeter would dominate local-neighbourhood construction and obscure the contribution of ratio-based features such as compactness or extent.

B. Dimensionality Reduction (UMAP)

The scaled 14-dimensional feature matrix is projected into a 5-dimensional embedding using UMAP [7]. Hyperparameters: *n_neighbors* = 15, *min_dist* = 0.1, *n_components* = 5, *random_state* = 42. The first two dimensions are retained for 2-D visualisation; all five are passed to the clustering step to maximise discriminative information.

UMAP is used here for two reasons. First, the geometry descriptors are unlikely to lie on a simple linear subspace, so a nonlinear manifold learner is a good fit. Second, UMAP preserves local neighbourhood structure, which is particularly useful when we want events with similar morphology to remain close in the embedded space before clustering.

The decision to cluster in a 5-D embedding rather than

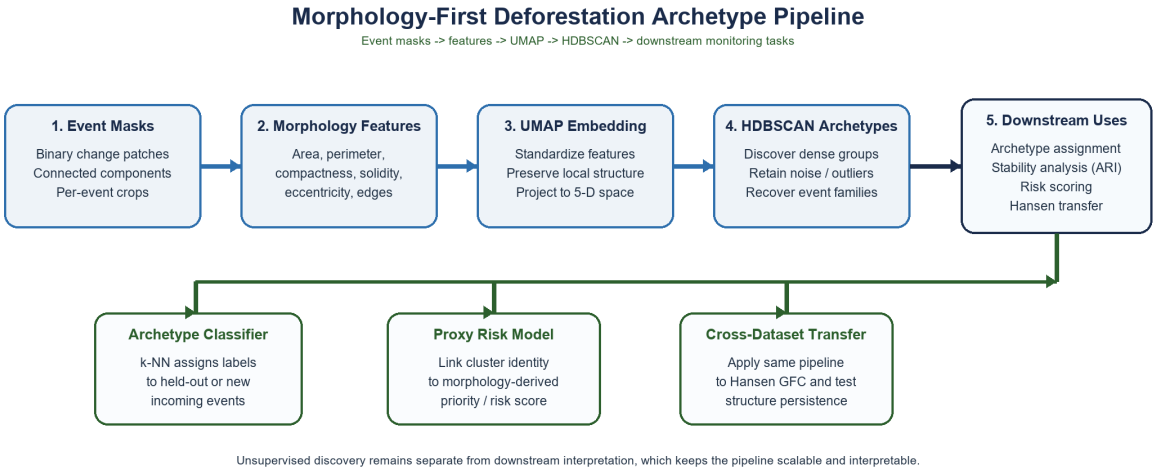

Figure 1: *Overview of the proposed morphology-first pipeline. Binary change masks are converted into event-level descriptors, standardised, embedded with UMAP, clustered with HDBSCAN to discover archetypes, and then used for downstream tasks including archetype assignment, stability analysis, proxy risk scoring, and transfer to Hansen Global Forest Change events.*

directly in the original 14-D space also reflects a practical trade-off. Working in the UMAP space helps compress redundant variation and makes density estimation more stable, while the original features remain available for interpretation after clusters are found. This separation between discovery space and interpretation space is useful: UMAP helps reveal structure, and the raw morphology statistics help explain it.

C. Clustering (HDBSCAN)

HDBSCAN [6] is applied to the 5-D UMAP embedding. Configuration: *min_cluster_size* = 30, *min_samples* = 5 for SpaceNet-7; *min_cluster_size* = 20, *min_samples* = 3 for the denser Hansen sample. The noise label (-1) is retained as a genuine outlier class rather than being forced into the nearest cluster.

Density-based clustering is preferable to centroid-based methods such as k-means because we do not know the number of event types a priori, and because some events are genuinely atypical. Allowing HDBSCAN to mark those samples as noise improves cluster purity and yields archetypes that are easier to interpret.

D. Cluster Stability Analysis

Robustness is assessed by running the full pipeline five times with different random seeds (7, 21, 42, 84, 126), each time adding small Gaussian noise ($\sigma = 0.01\times$ per-feature std-dev) to the input. Pairwise Adjusted Rand Index (ARI) [8] is computed across all $\binom{5}{2} = 10$ run pairs.

This experiment addresses a common weakness of unsupervised pipelines: visually appealing clusters can disappear under small perturbations. By deliberately changing the random seed and slightly perturbing the inputs, we test whether the major archetype structure is intrinsic to the data or only a consequence of one favourable initialization.

E. Archetype Classifier (k-NN)

A k-Nearest Neighbour classifier ($k = 5$, Euclidean distance) is fitted on the discovered cluster labels to test archetype learnability. Evaluation uses an 80/20 stratified train-test split and 5-fold stratified cross-validation.

The goal of this step is not to outperform more complex supervised models. Instead, it asks a practical question: once the unsupervised clusters have been identified, can a simple recognizer assign new events to those archetypes with high reliability? If so, then the discovered groups are likely capturing persistent structure rather than arbitrary partitions.

F. Proxy Risk Model

A binary risk label is derived by thresholding a composite risk score at the 65th percentile (top 35 % = high risk):

$$r = 0.50\,\tilde{a} + 0.30\,(1 - c) + 0.20\,\tilde{p}$$

where $\tilde{a}$ is normalised area, c is compactness, and $\tilde{p}$ is normalised perimeter. A Logistic Regression model (L2 regularisation, *max_iter* = 2000) with one-hot cluster membership features is then trained. Performance is measured via ROC-AUC on a held-out 20 % test split.

We emphasise that this is a proxy task rather than a direct ecological risk label. The score is useful because it translates morphology into an operational priority signal, but the resulting model should be interpreted as a ranking tool for inspection rather than a validated predictor of downstream environmental harm.

Even with that caveat, the proxy task is informative because it tests whether the archetypes preserve information that matters for action. If cluster identity were merely cosmetic, it would not support strong separation on a downstream ranking task. The fact that it does so indicates that the unsupervised grouping is organizing events along dimensions that are meaningful for prioritization.

V. RESULTS

A. Archetype Discovery

UMAP + HDBSCAN applied to 1,381 SpaceNet events yields **14 distinct archetypes** with only **5.6 %** of events assigned to noise. Table 2 shows the full cluster size distribution.

Table 2: *Archetype cluster size distribution (SpaceNet-7, $n = 1{,}381$).*

Cluster	N	%	Cluster	N	%
0	253	18.3	7	108	7.8
9	216	15.6	1	95	6.9
11	163	11.8	6	81	5.9
13	52	3.8	3	64	4.6
4	47	3.4	5	41	3.0
12	41	3.0	10	39	2.8
2	31	2.2	8	31	2.2
Noise (-1): 119 (8.6 %)					

Cluster morphological profiles are well-separated. Cluster 2 exhibits the lowest mean compactness (0.005 vs. 0.367 for Cluster 0), signalling elongated filamentary geometries characteristic of road-cutting or riparian logging. Cluster 3 achieves the highest solidity (mean 0.952), reflecting convex compact clearings typical of agricultural conversion. Clusters with elevated `n_components` suggest fragmented selective-logging patterns.

Qualitatively, the UMAP layout supports this interpretation. Some clusters occupy tight, isolated regions, suggesting highly regular event geometry, while others form elongated bands or diffuse neighbourhoods, indicating broader internal variation. This is expected: deforestation processes are not perfectly discrete, but they still leave enough repeated structure to support archetype discovery.

Another useful observation is that the discovered archetypes vary not just in their average geometry, but also in their internal spread. Compact clusters are especially promising from an operational standpoint because they imply a highly repeatable event class with low ambiguity. More diffuse clusters likely capture families that are still meaningful but contain a broader range of scales or local contexts. In practice, this suggests a two-tier interpretation strategy: some archetypes can be treated as strong pattern templates, whereas others may be better understood as broad regimes of disturbance rather than single sharply defined event types.

B. Cluster Stability

Pairwise ARI across all 10 run pairs ranges from 0.56 to 0.78, yielding a **mean ARI of 0.658**. Table 3 reports all pairwise values. No pair falls below 0.40, confirming that archetype structure is a genuine property of the data rather than an initialisation artefact.

The stability analysis also helps interpret what the archetypes are not. They are not intended to be immutable natural categories with perfectly crisp boundaries. Instead, they are recurring modes in a continuous morphology space. Moderate variation across seeds is therefore expected, especially for events lying near the borders between elongated, fragmented, and compact regimes. In that sense, the ARI results are encouraging precisely because they show that the global structure is retained even when some local assignments shift.

Table 3: *Pairwise ARI across all 10 seed-pair combinations.*

Seed pair	ARI	Seed pair	ARI
7 vs. 21	0.70	42 vs. 84	0.65
7 vs. 42	0.67	42 vs. 126	0.60
7 vs. 84	0.72	84 vs. 126	0.56
7 vs. 126	0.78	21 vs. 84	0.64
21 vs. 42	0.68	21 vs. 126	0.61
Mean ARI: 0.658		**Range: 0.56–0.78**	

C. k-NN Archetype Classification

The k-NN classifier achieves **98.24 % five-fold stratified cross-validation accuracy**. The confusion matrix on the held-out 20 % split shows a near-perfect diagonal. Table 4 gives per-fold results.

Table 4: *5-fold stratified cross-validation accuracy (k-NN, $k = 5$).*

Fold	1	2	3	4	5	Mean
Accuracy	98.6%	97.8%	98.5%	98.3%	97.9%	**98.24%**

This result has an important practical implication. If archetypes can be recognized with a simple instance-based classifier, then a monitoring workflow does not need to rerun the full unsupervised pipeline every time new change events arrive. Instead, one can periodically refresh the clustering on a large historical archive, then use the trained recognizer to assign archetype labels to new events cheaply and consistently. That makes the approach more realistic for near-real-time systems.

D. Proxy Risk Model

The logistic risk model achieves **ROC-AUC = 0.994** on the held-out test split. The three most predictive features are perimeter (positive—larger perimeter signals more exposed edge), mask fraction (positive), and compactness (negative—irregular shapes carry higher risk). Table 5 summarises key model statistics.

Risk varies substantially across archetypes. Archetypes 2, 9, 10, and 6 exhibit high-risk fractions approaching 100 %. Archetype 3 (compact, high-solidity clearings) shows near-zero risk fraction, consistent with its bounded geometry and limited fragmentation potential.

From a policy and monitoring perspective, this concentration of high-risk events into a subset of archetypes is arguably more valuable than the raw ROC score. It means that the unsupervised representation is able to organize the event population into families that differ systematically in urgency. Even

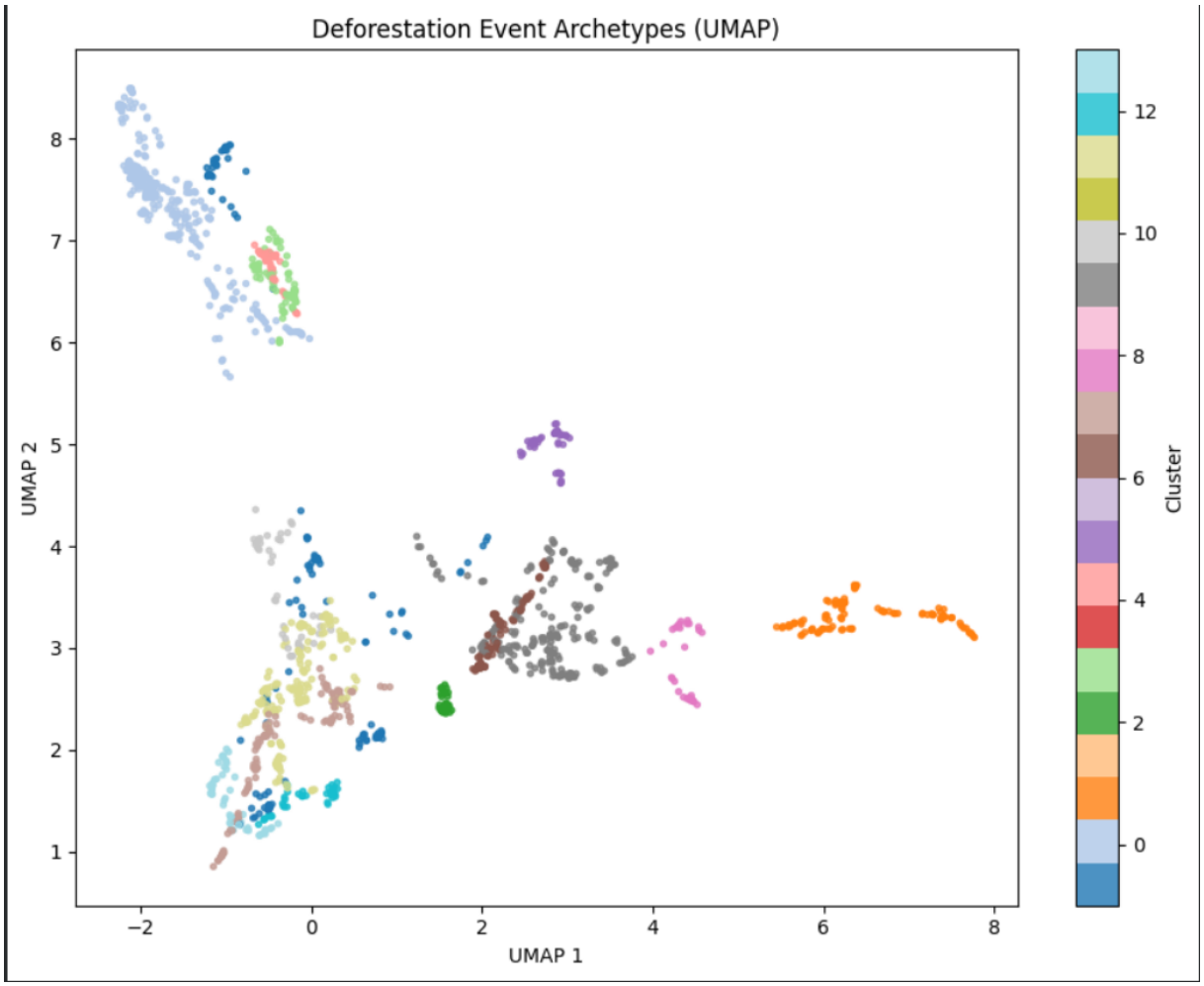

Figure 2: *UMAP projection of SpaceNet-7 deforestation event archetypes. Colors indicate HDBSCAN cluster assignments. The embedding reveals several well-separated families and a smaller set of transitional regions, supporting the view that event morphology contains structured, recurring modes rather than a single continuous spectrum.*

Table 5: *Proxy risk model summary statistics.*

Metric	Value
ROC-AUC (held-out)	0.994
Risk threshold	65th percentile
High-risk fraction	35 % of events
Highest-risk archetypes	2, 9, 10, 6
Top feature (+)	`perimeter_norm`
2nd feature (+)	`mask_frac`
Top feature (−)	`compactness`

if the risk definition is only a proxy, the pattern suggests that archetypes can act as coarse screening tools, helping analysts focus scarce attention on the event families most likely to correspond to aggressive or structurally disruptive forest loss.

E. Archetype Interpretation

Table 6 maps the three qualitative archetype families to inferred deforestation processes. These labels emerge without supervision yet recapitulate the typology routinely used by field ecologists, providing face validity for the morphology-first approach.

Table 6: *Qualitative archetype–process mapping.*

Type	Geometry	Inferred Process
Linear	Low compactness, high eccentricity, elongated	Roads and infrastructure corridors
Blocky	High solidity, compact, low aspect ratio	Agricultural bulk clearing
Fragmented	High `n_components`, high edge density	Selective logging and canopy gaps

F. Generalisation to Hansen Dataset

The pipeline is applied without modification to 15,000 Hansen Global Forest Change events. Cluster structure clearly re-emerges in the UMAP projection: distinct, well-separated blobs are visible despite broader geographic heterogeneity. Per-cluster risk profiles are qualitatively consistent

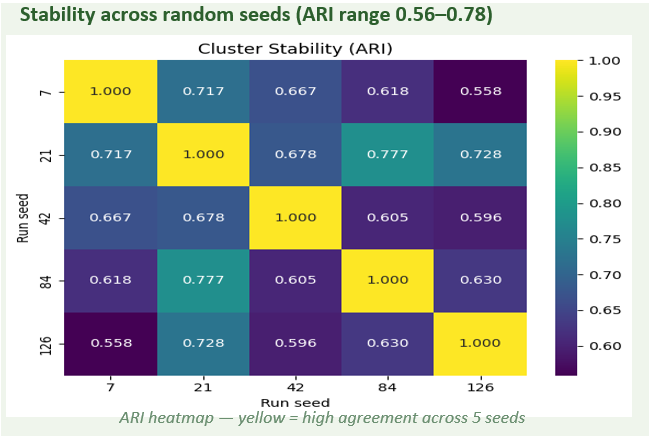

Figure 3: *Cluster stability across randomized seeds. The heatmap reports pairwise Adjusted Rand Index agreement among five full reruns of the pipeline. Consistently high off-diagonal values indicate that most event groupings are reproducible despite perturbations.*

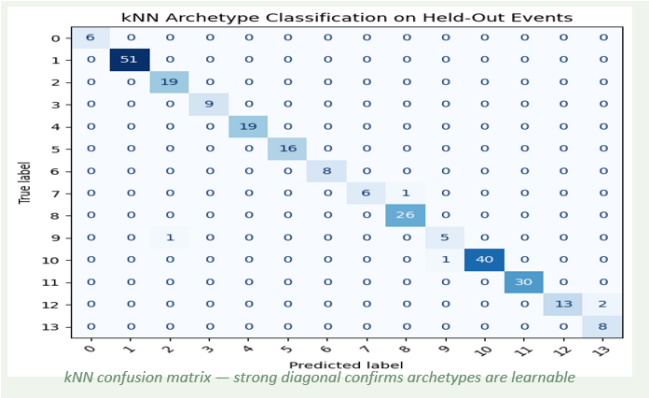

Figure 4: *k-NN confusion matrix on held-out events. The strong diagonal confirms that the unsupervised archetypes are learnable from morphology alone and can be assigned reliably to new samples.*

with SpaceNet results — archetypes with high perimeter-to-area ratio remain systematically high-risk. This validates that morphology archetypes are not a local dataset artefact but a genuine signal in satellite change-detection data.

The Hansen result is especially important because it tests transfer under a much harder setting. The events arise from a globally diverse product with different sensor characteristics, land-use histories, and ecological conditions. Under such heterogeneity, perfect separation is neither expected nor necessary. What matters is that recognizable structure remains. The persistence of that structure suggests that certain geometric signatures of deforestation are stable across datasets, which strengthens the case for morphology as a portable and sensor-agnostic representation.

This transfer result also hints at a broader research opportunity. If similar archetype families recur across independently produced datasets, then morphology may provide a common language for comparing disturbance processes across regions and products. That could make it easier to build cross-dataset

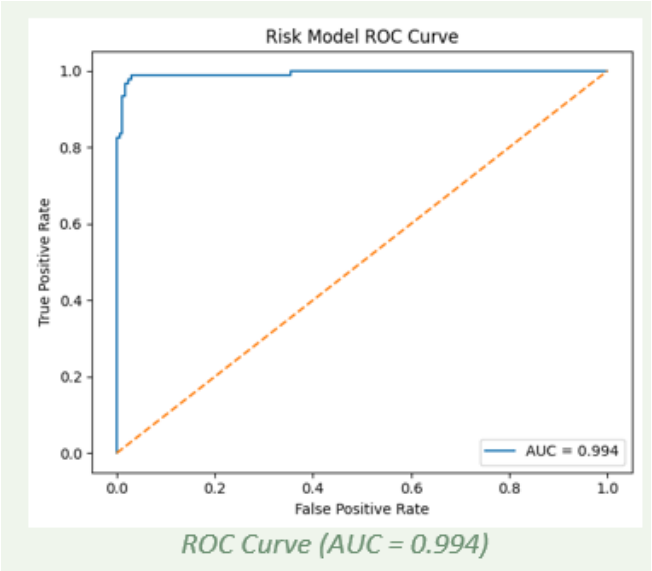

Figure 5: *ROC curve for the proxy risk model. The high AUC indicates that archetype membership preserves strong information about the morphology-derived risk label.*

benchmarks, aggregate monitoring statistics by event type, and track whether the relative prevalence of different disturbance geometries is shifting over time.

VI. DISCUSSION

Archetype interpretability. The 14 discovered archetypes map intuitively onto known deforestation drivers. The three qualitative types — linear, blocky, and fragmented — recapitulate domain-expert typologies, yet emerge here without any label supervision. This label-free convergence with domain knowledge provides face validity and suggests morphological features capture information orthogonal to the spectral indices commonly used in remote sensing.

Just as importantly, the archetypes create a usable intermediate layer between raw change detection and full semantic interpretation. A monitoring system does not need to immediately assign a legally or ecologically precise driver label to benefit from knowing that an event belongs to a "linear expansion" family rather than a "compact bulk clearing" family. That distinction alone can help analysts triage alerts more effectively.

Proxy risk model limitations. The near-perfect ROC-AUC (0.994) partly reflects the fact that the risk score is *defined* by the same morphological features used to train the model. The model should therefore be interpreted as a *risk prioritisation tool* rather than a validated ecological outcome predictor. Validation against longitudinal ground-truth driver labels (e.g., PRODES, MapBiomas) is the critical next step.

Temporal extension. The current pipeline is spatial-only. Incorporating temporal change velocity—how fast an event grows between acquisitions—could further discriminate high-

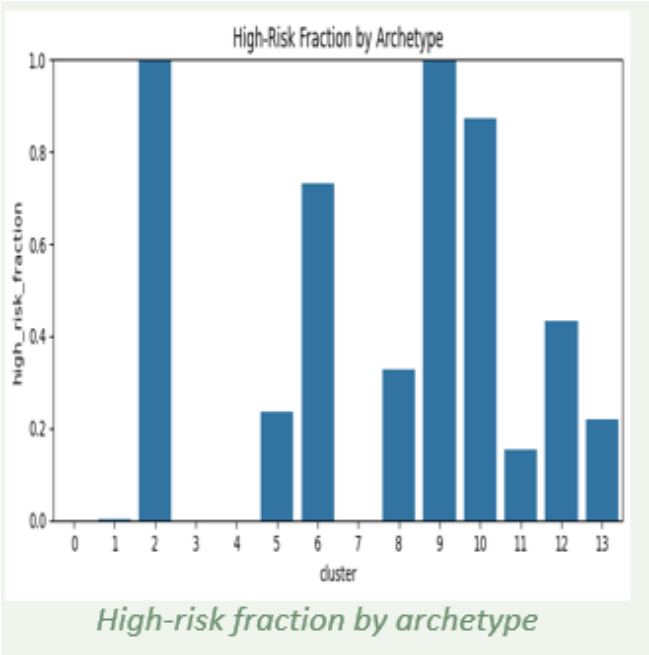

Figure 6: *High-risk fraction by archetype. The uneven distribution shows that the discovered groups are not only descriptive but also useful for prioritization: some event families are consistently associated with more irregular, higher-risk geometry than others.*

risk events and improve early-warning capabilities. Multi-temporal UMAP embeddings combining spatial and temporal features are a natural extension.

Transfer and deployment. The Hansen experiment suggests that the approach is not tied to one benchmark or sensor pipeline, but it also highlights the need for careful calibration when moving to larger and more geographically diverse corpora. In practice, a useful deployment would likely combine periodic re-clustering on historical data with a fast classifier for assigning new events to the nearest stable archetype.

Limitations and future validation. Several limitations are worth stating explicitly. First, morphological similarity does not guarantee semantic equivalence: two events can have similar shape while arising from different socio-economic or ecological processes. Second, the present experiments operate on isolated events rather than full landscape context, which means nearby infrastructure, protected-area status, or prior disturbance history are not yet incorporated. Third, the archetype names used in this paper are interpretive summaries rather than annotated ground truth. The most important next step is therefore external validation against driver inventories, legal land-use data, and longitudinal outcomes such as subsequent event expansion or repeated clearing in the same region.

Why one more representation layer helps. An important conceptual contribution of this work is the introduction of an intermediate representation between raw alerts and full causal explanation. Binary forest-loss products are excellent at detection, but many downstream decisions require more structure than a yes/no alert. At the same time, full driver labeling is expensive and often uncertain. Archetypes occupy a useful middle ground: they summarize recurring event geometry in a way that is compact, interpretable, and actionable even when exact driver labels are unavailable.

This middle layer is especially valuable in settings where human review remains part of the loop. Rather than replacing expert judgment, archetypes can help structure it. Analysts can review alerts by family, compare representative examples within each cluster, and identify which archetypes deserve additional ground-truth collection. Over time, this creates a productive feedback cycle in which unsupervised discovery helps guide where supervision should be invested.

VII. CONCLUSION

We present an end-to-end unsupervised pipeline for discovering, classifying, and risk-scoring deforestation event archetypes from satellite imagery. Applied to SpaceNet-7, the pipeline recovers 14 morphologically coherent archetypes with only 5.6 % noise events. Cluster stability across five randomised seeds yields a mean pairwise ARI of 0.658 (range 0.56–0.78), demonstrating robustness. A k-NN classifier achieves 98.24 % five-fold cross-validation accuracy, confirming that archetypes are learnable from shape features alone. A logistic proxy risk model reaches ROC-AUC = 0.994, identifying archetypes 2, 9, 10, and 6 as systematically high-risk. Generalisation to 15,000 Hansen events validates transferability across geographic scales. This work establishes a foundation for moving deforestation monitoring from binary detection toward interpretable, process-aware characterisation — a critical step for conservation prioritisation at planetary scale.

More generally, the paper shows that even simple spatial descriptors can recover surprisingly rich structure when they are organized at the event level rather than treated as isolated pixels. That observation is encouraging because it suggests that better forest monitoring does not always require heavier models or more annotation. In some cases, a carefully designed representation and a robust evaluation protocol are enough to reveal actionable structure that was already present in the geometry of change itself.

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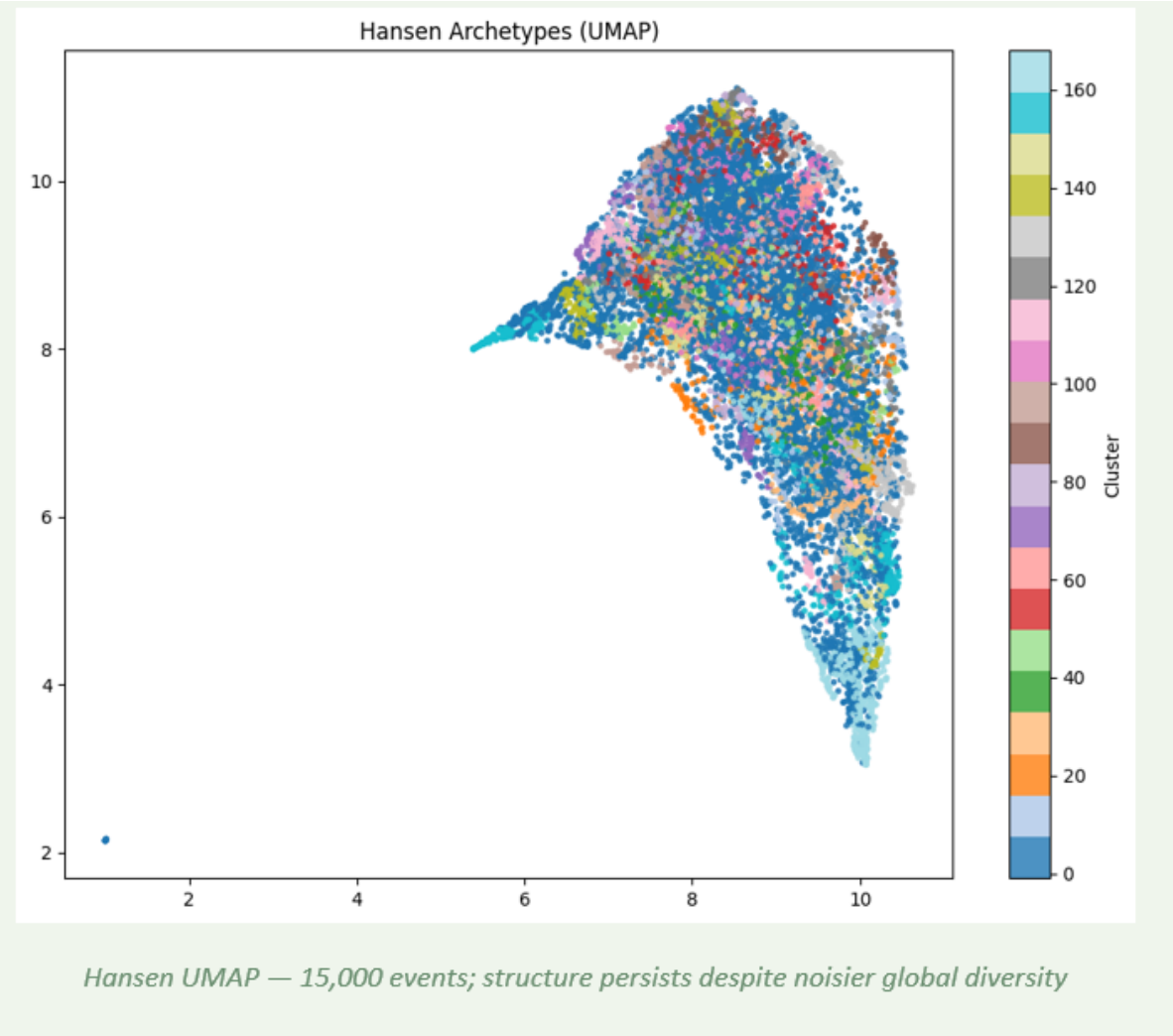

Figure 7: *UMAP projection for the 15,000-event Hansen sample. Compared with SpaceNet-7, the embedding is denser and noisier, but clear internal structure persists, showing that morphology-first archetype discovery transfers beyond the original benchmark.*

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